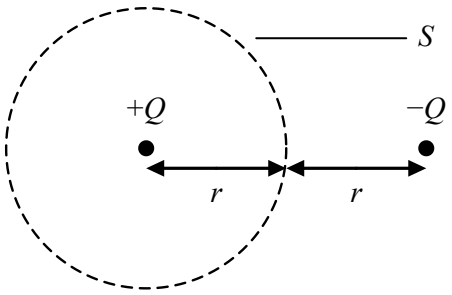
STPM 2020 PHYSICS 960/2 (U2)

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. Two point charges, $+Q$ and $-Q$, are $2r$ apart in vacuum as shown in the diagram below.



If $+Q$ is positioned at the centre of the spherical surface, S, with the radius of r, what is the electric flux through S?

A 0 B $\frac{+Q}{8\pi\varepsilon_o r}$ C $\frac{+Q}{4\pi\varepsilon_o r}$ D $\frac{+Q}{\varepsilon_o}$

Answer: D

Only $+Q$ is enclosed by S: $\Phi = \frac{+Q}{\varepsilon_o}$

Explanation: Charges outside the surface contribute zero net flux.

2. A Van de Graaff generator has a dome of radius 20.0 cm. The insulation of air breaks down when the electric field exceeds 3.0×10^6 V m^{-1}. What is the maximum electric potential and maximum charge on the dome?

	Maximum electric potential	*Maximum charge*
A	6.0×10^5 V	6.7×10^{-5} C
B	6.0×10^5 V	1.3×10^{-5} C
C	1.5×10^7 V	6.7×10^{-5} C
D	1.5×10^7 V	1.3×10^{-5} C

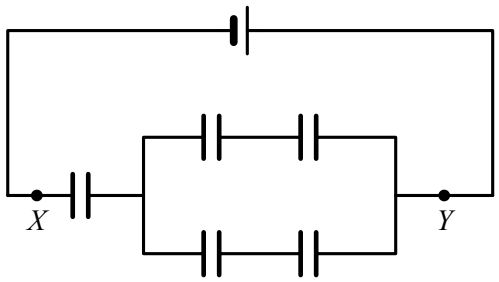
Answer: B

$$V = Er = (3.0 \times 10^6)(0.200) = 6.0 \times 10^5 \text{ V}$$

$$Q = \frac{Er^2}{k} = \frac{(3.0 \times 10^6)(0.200)^2}{9.0 \times 10^9} = 1.3 \times 10^{-5} \text{ C}$$

Explanation: At the surface $E = kQ/r^2$ and $V = kQ/r$, so $V = Er$.

3. Five identical capacitors are connected to a battery as shown in the circuit diagram below.



If the capacitance of each capacitor is 20 μF, what is the effective capacitance between *X* and *Y*?

A 10 μF B 20 μF C 30 μF D 40 μF

Answer: A

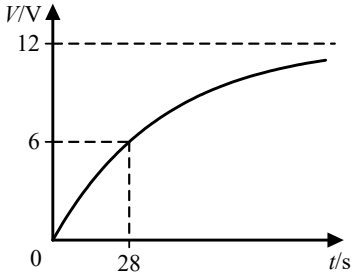
Each branch: two 20 μF in series = 10 μF; two branches in parallel = 20 μF.

$$\text{In series with 20 μF: } \frac{(20)(20)}{20+20} = 10 \text{ μF}$$

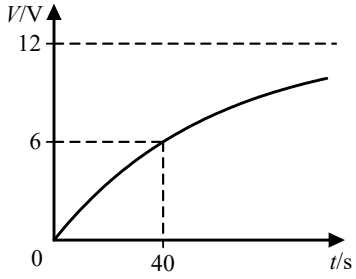
Explanation: Work from the innermost group outwards.

4. When a capacitor of capacitance 20 μF is charged by a battery of e.m.f. 12 V through a resistor, the initial current is 6.0 μA. Which graph represents the variation of the potential difference, *V*, across the capacitor with time, *t*?

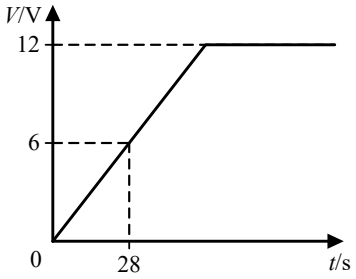
A



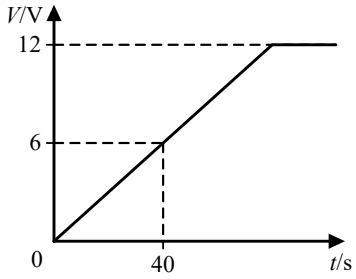
B



C



D



Answer: A

$$R = \frac{12}{6.0 \times 10^{-6}} = 2.0 \times 10^{6}\ \Omega;\ \tau = RC = (2.0 \times 10^{6})(20 \times 10^{-6}) = 40\ \text{s}$$

$V = 6$ V (half of 12 V) at $t = RC \ln 2 = 28$ s, on an exponential curve (A)

Explanation: The 40 s is the time constant, when V = 63% of 12 V.

5. The differences of physical quantities between electrical conduction and thermal conduction in a metal conductor are given in the table below.

Electrical conduction	**Thermal conduction**
Electrical conductivity	Thermal conductivity
Electrical charges	X
Y	Rate of heat flow

Which physical quantities represent X and Y?

	X	Y
A	Temperature difference	Cross-sectional area
B	Cross-sectional area	Length of conductor
C	Heat	Current
D	Length of conductor	Potential difference

Answer: C

Electrical charge corresponds to heat (X), and the rate of heat flow corresponds to the rate of flow of charge, i.e. current (Y).

Explanation: Potential difference corresponds to temperature difference.

6. A heating coil with a power rating of 2.00 A, 500 W is used in a house. If the resistivity of the coil is 1.00×10^{-6} Ω m and its diameter is 0.40 mm, what is the length of the coil?

A 15.7 m B 31.4 m C 62.8 m D 125 m

Answer: A

$$R = \frac{P}{I^2} = \frac{500}{2.00^2} = 125\ \Omega;\ A = \pi(0.20 \times 10^{-3})^2 = 1.26 \times 10^{-7}\ \text{m}^2$$

$$l = \frac{RA}{\rho} = \frac{(125)(1.26 \times 10^{-7})}{1.00 \times 10^{-6}} = 15.7\ \text{m}$$

Explanation: Use the radius, half the diameter.

7. Which statement is true for internal resistance of a cell in a closed circuit?

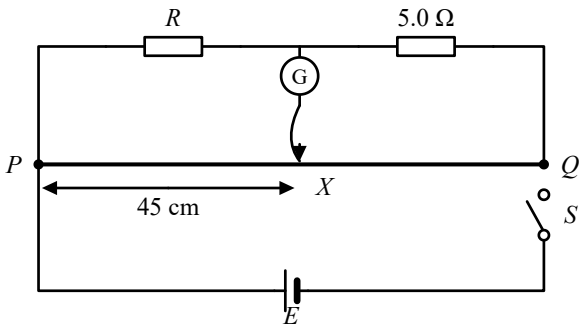
A The greater the internal resistance of a cell the smaller the e.m.f.
B Internal resistance of a cell is the ratio of the e.m.f. to the current.
C The greater the internal resistance of a cell the smaller the terminal potential difference.
D Internal resistance of a cell is the ratio of the terminal potential difference to the current.

Answer: C

$V = \varepsilon - Ir$: a greater internal resistance gives a smaller terminal p.d. (C)

Explanation: The e.m.f. does not depend on r.

8. A resistor of resistance 5.0 Ω and another resistor, R, are connected in a Wheatstone bridge as shown in the circuit diagram below.



The length of the slide wire PQ is 100 cm. When the switch, S, is closed, the balanced length PX is 45 cm. What is the resistance of R?

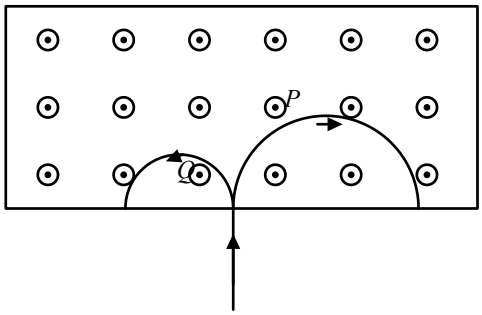
A 2.3 Ω B 4.1 Ω C 6.1 Ω D 11 Ω

Answer: B

$$\frac{R}{5.0} = \frac{45}{55} \Longrightarrow R = 4.1\ \Omega$$

Explanation: At balance the ratio of resistances equals the ratio of wire lengths.

9. Two charged particles, P and Q, carrying an equal magnitude of charges and travel with the same velocity, enter a uniform magnetic field pointing out of the page. The paths of the particles after entering the magnetic field are shown in the diagram below.



What are the sign of charges for P and Q, and the relation between the mass of the particles?

	Sign of charge for P	*Sign of charge for Q*	*Relation between the mass*
A	Positive	Negative	$P > Q$
B	Negative	Positive	$P < Q$
C	Positive	Negative	$P < Q$
D	Negative	Positive	$P > Q$

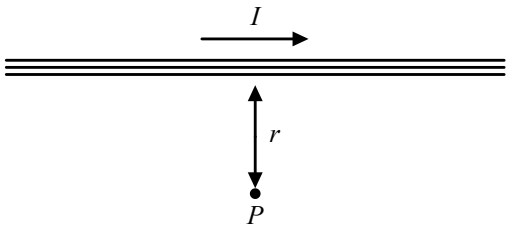
Answer: A

Moving up with **B** out of the page, a positive charge is pushed to the right: P is positive and Q is negative.

$$r = \frac{mv}{qB} \propto m\text{: } P \text{ has the larger radius, so } P > Q \text{ in mass}$$

Explanation: Same charge magnitude and speed, so the radius depends only on the mass.

10. A bundle of three long and thin straight wires each carries current, I, flows in the same direction as shown in the diagram below.



The magnetic flux density at P at distance r from the wires is

A $\frac{\mu_o I}{2\pi r}$ into the paper

B $\frac{\mu_o I}{2\pi r}$ out of the paper

C $\frac{3\mu_o I}{2\pi r}$ into the paper

D $\frac{3\mu_o I}{2\pi r}$ out of the paper

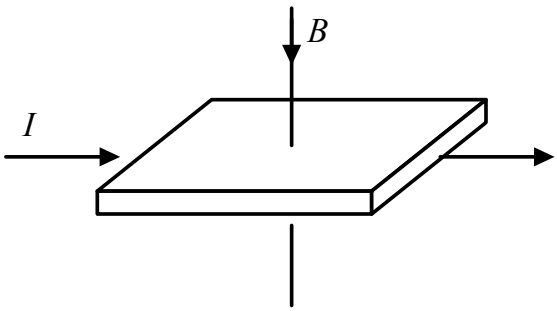
Answer: C

The three wires act as one wire carrying $3I$.

$B = \frac{\mu_o(3I)}{2\pi r}$; below a current flowing to the right, **B** is into the paper

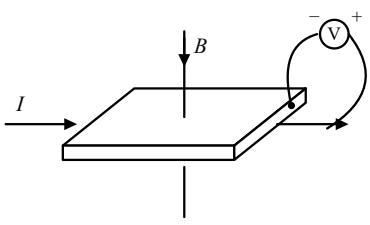
Explanation: Use the right-hand grip rule.

11. A current, I, flows through a piece of gold foil placed in a magnetic field, B, is as shown in the diagram below.

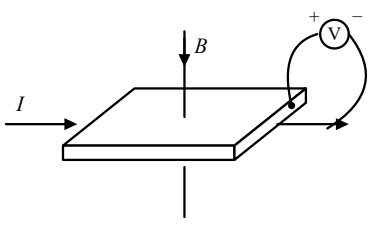


A voltmeter is used to measure the Hall voltage. Which diagram shows the connection with the correct polarity of the voltmeter?

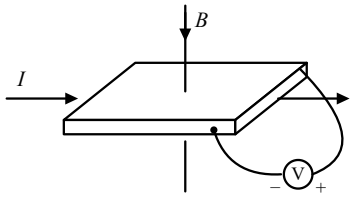
A



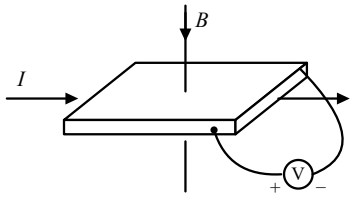
B



C



D



Answer: D

In gold the carriers are electrons, moving opposite to *I*. The magnetic force pushes them to the far (back) face, which becomes negative; the near (front) face is positive.

The voltmeter's + terminal must connect to the front face and − to the back face (D).

Explanation: Using $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ with $q = -e$.

12. An e.m.f. of 0.60 V is induced in a coil when the current decreases from 0.50 A to zero in 0.40 s. What is the self inductance of the coil?

A 0.33 H B 0.48 H C 0.75 H D 1.25 H

Answer: B

$$L = \frac{\varepsilon}{\mathrm{d}I/\mathrm{d}t} = \frac{0.60}{0.50/0.40} = 0.48 \text{ H}$$

Explanation: The rate of change of current is 1.25 A s^{-1}.

13. The mutual inductance between two nearby coils is one henry if

A an e.m.f. of one volt is induced in the first coil when the current of one ampere is flowing in the second coil

B an e.m.f. of one volt is induced in the first coil when the current in the second coil is changing at one ampere per second

C a current of one ampere per second is induced in the first coil when the potential difference of one volt is applied to the second coil

D a current of one ampere is induced in the first coil when the potential difference at the second coil is changing at one volt per second

Answer: B

M = 1 H when a current changing at 1 A s^{-1} in one coil induces an e.m.f. of 1 V in the other (B).

Explanation: $\varepsilon_1 = -M\,\mathrm{d}I_2/\mathrm{d}t$.

14. When a sinusoidal voltage is applied to an ideal capacitor, the average power is always zero because

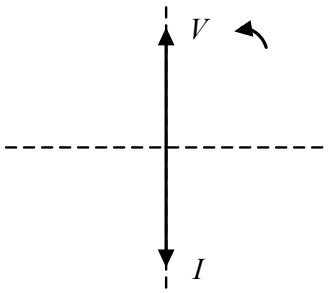
A the instantaneous power of the capacitor is always zero
B the power dissipated by the capacitor is always positive
C the energy stored in the capacitor during charging is equal to the heat dissipated in the capacitor during discharging
D the energy stored in the capacitor during charging is equal to the energy returned to the supply during discharging

Answer: D
Energy stored in the field while the capacitor charges is returned to the supply as it discharges, so the average power is zero (D).

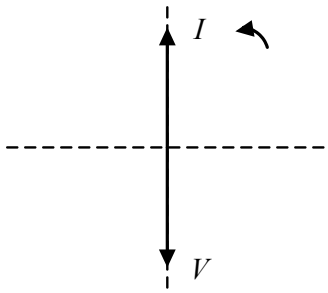
Explanation: An ideal capacitor dissipates no heat.

15. Which phasor diagram shows the correct relationship between voltage, V, and current, I, when a capacitor is connected across a sinusoidal a.c. supply?

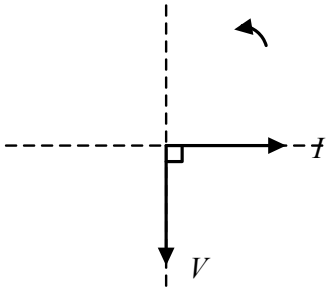
A



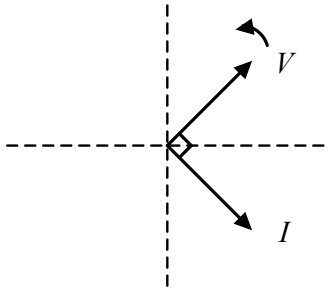
B



C



D

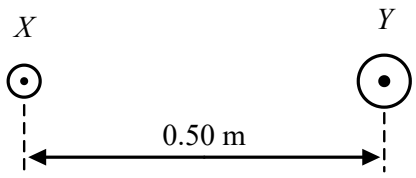


Answer: C
In a capacitor the current leads the voltage by 90°. With anticlockwise rotation, I must be 90° ahead of V: V downwards and I to the right (C).

Explanation: Remember "ICE": in a capacitor (C), current (I) leads e.m.f. (E).

Section B [15 marks]

16. A conducting sphere, *X*, of radius 0.15 cm is fixed at a distance of 0.50 m from a conducting sphere, *Y*, of radius 0.30 cm as shown in the diagram below.



The charges on *X* and *Y* are +3.0 μC and −2.0 μC respectively.

(a) Determine the force between *X* and *Y*. [2]

Answer

$F = \dfrac{kQ_X Q_Y}{r^2} = \dfrac{(9.0 \times 10^9)(3.0 \times 10^{-6})(2.0 \times 10^{-6})}{(0.50)^2}$ — 1

$F = 0.216$ N, attractive — 1

2

Explanation: The spheres are tiny compared with their separation, so treat them as point charges.

(b)

(i) Explain what happens to the charges when *X* and *Y* are connected by a thin wire. [3]

Answer

The spheres are at different potentials, so charge flows through the wire (electrons move from *Y* to *X*) — 1

The flow stops when both spheres reach the same potential — 1

The total charge, +3.0 − 2.0 = +1.0 μC, is conserved and shared between them — 1

3

Explanation: Electrons flow from lower to higher potential.

(ii) The thin wire is then removed. Determine the final charge on *Y*. [3]

Answer

Same potential: $\dfrac{kq_X}{r_X} = \dfrac{kq_Y}{r_Y} \Rightarrow q_Y = 2q_X$ — 1

$q_X + q_Y = 1.0$ μC — 1

$q_Y = \dfrac{2}{3}(1.0\ \mu\text{C}) = 0.67\ \mu\text{C}$ — 1

3

Total for question **8**

Explanation: Charge shares in proportion to radius.

17. A resistor of resistance 50.0 Ω and an inductor of inductance 0.200 H are connected in series with a sinusoidal source, $V = 240.0 \sin (120t)$.

(a) State the phase relation between current and voltage across the inductor. [2]

Answer

The voltage across the inductor leads the current 1

by 90° ($\frac{\pi}{2}$ rad) 1

2

Explanation: Equivalently, the current lags the voltage by 90°.

(b) Calculate the impedance of the circuit. [3]

Answer

$X_L = \omega L = (120)(0.200) = 24.0\ \Omega$ 1

$Z = \sqrt{(50.0^2 + 24.0^2)}$ 1

$Z = 55.5\ \Omega$ 1

3

Explanation: $\omega = 120$ rad s^{-1} is read from the equation.

(c) Determine the maximum voltage across the inductor. [2]

Answer

$I_0 = \frac{240.0}{55.5} = 4.33$ A 1

$V_{L,0} = I_0 X_L = (4.33)(24.0) = 104$ V 1

2

Total for question **7**

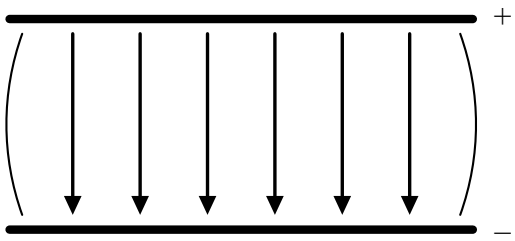
Explanation: The peaks of V_R and V_L do not occur at the same time.

Section C [30 marks]

18. (a) A capacitor consists of two square parallel plates separated at a distance 1.0 mm. Each sides of the plates is 5.0 cm. The magnitude of the electric field between the plates is 5.00×10^4 N C^{-1} and the potential difference between the plates is maintained.

(i) Sketch a diagram to show the electric field lines in the region between the plates. [2]

Answer



uniform, equally spaced lines from + to −; slight bulging at the edges

Parallel, equally spaced straight lines between the plates	1
Arrows from the positive to the negative plate (edge effect optional)	1
	2

Explanation: Equal spacing shows the field is uniform.

(ii) Calculate the potential difference between the plates. [2]

Answer

$V = Ed = (5.00 \times 10^4)(1.0 \times 10^{-3})$	1
$V = 50$ V	1
	2

Explanation: N C^{-1} is the same as V m^{-1}.

(iii) Calculate the energy stored in the capacitor. [4]

Answer

$A = (0.050)^2 = 2.5 \times 10^{-3}\ \text{m}^2$ 1

$C = \frac{\varepsilon_0 A}{d} = \frac{(8.85 \times 10^{-12})(2.5 \times 10^{-3})}{1.0 \times 10^{-3}} = 2.21 \times 10^{-11}\ \text{F}$ 1

$U = \frac{1}{2}CV^2 = \frac{1}{2}(2.21 \times 10^{-11})(50)^2$ 1

$U = 2.77 \times 10^{-8}\ \text{J}$ 1

4

Explanation: Alternatively, energy density $\frac{1}{2}\varepsilon_0 E^2$ × volume gives the same.

(b) A capacitor is fully charged by a battery and then the connection is disconnected. A dielectric is inserted between the plates of the capacitor. Explain the effect of the dielectric on

(i) the capacitance of the capacitor [5]

Answer

The capacitor is isolated, so the charge Q is constant 1

The dielectric molecules are polarised by the field 1

Induced surface charges on the dielectric set up a field opposing the original field 1

The resultant field, and hence the p.d. $V = Ed$, decreases 1

$C = \frac{Q}{V}$ increases (by the factor ε_r) 1

5

Explanation: The capacitance depends on the material between the plates.

(ii) the energy stored in the capacitor [2]

Answer

$U = \frac{Q^2}{2C}$ with Q constant and C increased 1

The energy stored decreases (by the factor ε_r); the dielectric is pulled in, so the field does work on it 1

2

Total for question **15**

Explanation: With the battery connected instead, the energy would increase.

19. (a) Based on the motion of free electrons, describe

(i) the mechanism of electrical conduction in a metal conductor [2]

Answer

The applied p.d. sets up an electric field that makes the free electrons drift opposite to the field	1
This drift, superimposed on their random motion, is a net flow of charge: the current	1
	2

Explanation: The drift velocity is very small compared with the random speeds.

(ii) the heating effect in a metal conductor [2]

Answer

Between collisions the electrons gain kinetic energy from the field	1
In collisions they give this energy to the lattice ions, which vibrate more: the conductor gets hotter	1
	2

Explanation: This is Joule heating, I^2R.

(b) Show that the current density, $J = \sigma E$, is equivalent to Ohm's law where σ is the electrical conductivity and E is the electric field. [5]

Answer

$J = \frac{I}{A}$	1
$E = \frac{V}{l}$ and $\sigma = \frac{1}{\rho}$	1
$\frac{I}{A} = \frac{V}{\rho l}$	1
$V = I(\frac{\rho l}{A}) = IR$	1
For a given conductor at constant temperature, R is constant, so $V \propto I$: Ohm's law	1
	5

Explanation: $J = \sigma E$ is the microscopic form of Ohm's law.

(c) A potential difference of 0.30 V is applied across a wire of length 1.10 m and cross-sectional area 1.5×10^{-6} m^2. The wire carries a current of 4.0 A and the density of free electrons is 1.4×10^{29} m^{-3}. Determine the mean time between collisions of the free electrons. [6]

Answer

$J = \dfrac{4.0}{1.5 \times 10^{-6}} = 2.67 \times 10^6$ A m^{-2} 1

$E = \dfrac{0.30}{1.10} = 0.273$ V m^{-1} 1

$\sigma = \dfrac{J}{E} = 9.78 \times 10^6\ \Omega^{-1}$ m^{-1} 1

$\sigma = \dfrac{ne^2\tau}{m} \Longrightarrow \tau = \dfrac{m\sigma}{ne^2}$ 1

$\tau = \dfrac{(9.11 \times 10^{-31})(9.78 \times 10^6)}{(1.4 \times 10^{29})(1.60 \times 10^{-19})^2}$ 1

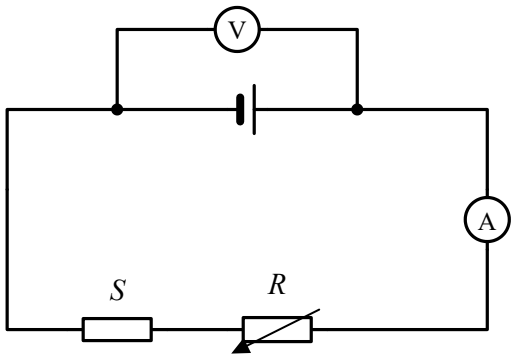
$\tau = 2.5 \times 10^{-15}$ s 1

6

Total for question **15**

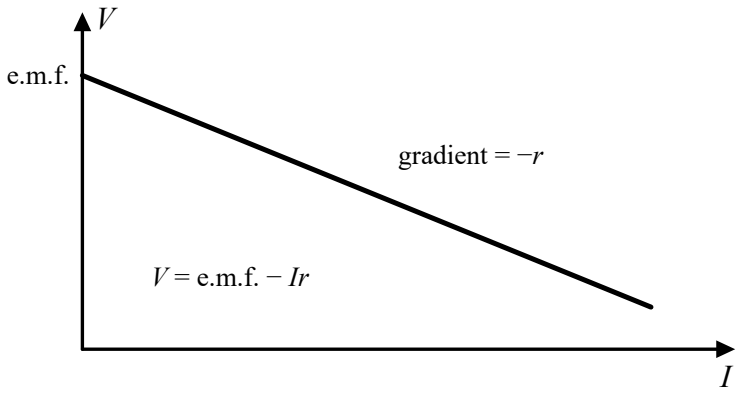
Explanation: $\sigma = ne^2\tau/m$ comes from $v_d = eE\tau/m$.

20. (a) A battery, a resistor, *S*, a variable resistor, *R*, an ammeter and a voltmeter are connected as shown in the circuit diagram below.



(i) Explain, with the aid of a suitable graph, the variation of the terminal potential difference with the current when R is increased gradually. [5]

Answer



Straight line with negative gradient	1
Intercept on the V-axis equal to the e.m.f.	1
Terminal p.d. $V = \varepsilon - Ir$	1
As R increases, the current decreases	1
so the lost volts Ir decrease and the terminal p.d. increases towards ε	1
	5

Explanation: Each setting of R gives one point (I, V) on the line.

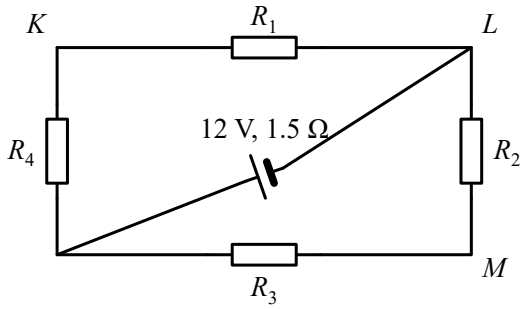
(ii) State how the value of internal resistance and e.m.f. of the battery could be obtained from the graph in (a)(i). [2]

Answer

E.m.f. = intercept on the V-axis (where $I = 0$)	1
Internal resistance = magnitude of the gradient	1
	2

Explanation: The intercept on the I-axis is the short-circuit current ε/r.

(b) Four resistors, R_1, R_2, R_3 and R_4 are connected to a battery of e.m.f. 12 V with internal resistance of 1.5 Ω as shown in the circuit diagram below.



The resistance of R_1, R_2, R_3 and R_4 are 10 Ω, 5.0 Ω, 15 Ω and 10 Ω respectively.

(i) Calculate the potential difference between K and L. [2]

Answer

Two paths from L to the other battery terminal: $R_1 + R_4 = 20$ Ω and $R_2 + R_3 = 20$ Ω, in parallel = 10 Ω; $I = \frac{12}{10 + 1.5} = 1.04$ A, 0.522 A in each path — 1

$V_{KL} = (0.522)(10) = 5.22$ V — 1

2

Explanation: The two paths are equal, so the current splits equally.

(ii) Determine the power which is supplied by the battery. [4]

Answer

External resistance = 10 Ω — 1

$I = \frac{\varepsilon}{R + r} = \frac{12}{11.5}$ — 1

$I = 1.04$ A — 1

$P = \varepsilon I = (12)(1.04) = 12.5$ W — 1

4

Explanation: Power supplied by the battery includes the power lost in its internal resistance.

(iii) If R_1 is replaced by an unknown resistor, R_x, determine the value of R_x so that the potential difference between K and M is zero. [2]

Answer

Balanced bridge: $\frac{R_x}{R_4} = \frac{R_2}{R_3} \Rightarrow \frac{R_x}{10} = \frac{5.0}{15}$ 1

$R_x = 3.33\ \Omega$ 1

2

Total for question **15**

Explanation: K and M are then at the same potential.